Dr. Vinnarasi Saravanan

Researcher in Computational Biophysics

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Summary

Computational biophysics researcher with a strong background in molecular simulations and free-energy methods for understanding biomolecular structure, dynamics, and interactions. My research focuses on using computational approaches to address biologically and therapeutically relevant questions, including protein–ligand interactions, DNA repair mechanisms, and the design of less immunogenic biomolecules through L-to-D amino acid substitutions. I am particularly interested in connecting molecular level insights from simulations with applications in drug discovery, therapeutic design, and biomedical research.

Education

- June 2008 – Apr-2009 — **SSLC (88%),** St. John's Girls Higher Secondary School, Devakottai, India.
- June 2010 – Apr-2011 — **HSC (75%),** St. John's Girls Higher Secondary School, Devakottai, India.
- Jun-2011 – Apr-2014 — **B.Sc. Physics (84%),** Ananda College, Devakottai, India
- Jun-2014 – Apr-2016 — **M.Sc. Physics (78%),** St.Joseph's College, Trichy, India
- Nov-2016 – Nov-2020 — **Ph.D.,** Department of Physics, Bharathiar University, India.
  Thesis title: *Theoretical Investigation of G-Quadruplex DNA with anti-cancer drugs for targeted drug delivery and two-dimensional nanomaterials for DNA biosensors.*

Research Experience

- Feb-2025 – Aug 2026 — **Postdoctoral Researcher,** I2BC, CEA-Paris Saclay University, France.
- Jun-2023 – Aug 2024 — **Postdoctoral Researcher,** UGSF, Université de Lille, France.
- Feb-2021 – Feb-2023 — **Postdoctoral Researcher,** Department of Chemical Engineering, Indian Institute of Technology Bombay, India
- Jun-2021 – Nov-2021 — **Teaching Assistant**, Numerical Methods, Undergraduate Programme (B.Tech.,), Department of Chemical Engineering, Indian Institute of Technology Bombay, India.

Technical Skills

- **Research Expertise:** All-atom and coarse- grained molecular dynamics simulations, Enhanced sampling techniques, Free energy calculations (FEP/TI), Molecular docking and structure-based drug design (SBDD), Quantum chemical calculations
- **Structure Modeling & Prediction:** AlphaFold, Homology Modeling, Loop Refinement, Mutant Modeling
- **Molecular Modeling & Quantum Tools:** GROMACS, NAMD, Gaussian 16, AutoDock
- **Visualization & Analysis Tools:** VMD, PyMOL, Chimera
- **Programming & Data Analysis:** Python, Jupyter Notebook, TCL
- **Research Interests:** Computational Biophysics, Therapeutic Biomolecular Design, DNA Repair Mechanisms.

Conference Presentations

2026 UGM and Conference 2026, Basel, Switzerland **(Poster presentation)**.

2024 Quatrième réunion annuelle du GDR Architecture et Dynamique du Noyau et des Génomes (ADN and G), Toulose, France **(Oral presentation)**.

3rd MOSBRI scientific conference – Ljubljana, Slovenia **(Poster presentation)**.

2022 1st MOSBRI2022 scientific conference- Pasteur Institute, Paris, France **(Poster presentation)**.

2019 Biomolecular structure and dynamics- Periyar University, India **(Oral presentation)**.

Theoretical Chemistry Symposium – BITS Pilani, India **(Poster presentation)**.

Publications

1. Saravanan, V., Raouraoua, N., Brysbaert, G., Giordano, S., Lensink, M., Cleri, F., Blossey, R. *The "very moment" when UDG recognizes a flipped-out uracil base in dsDNA*, *Scientific Reports*, 15, 7993 (2025). https://doi.org/10.1038/s41598-025-91705-6

2. Ghediri, S., Sarma, PAP., **Saravanan, V.**, Abbadie, C., Blossey, R., Cleri, F. *Mechanisms of DNA Damage Recognition by UDG and PARP1 in the Nucleosome*. *Biomolecules*, 15(5), 649, (2025). https://doi.org/10.3390/biom15050649

3. Saravanan, V., Ahammed, I., Bhattacharya, A., Bhattacharya, S. *Uncovering allostery and regulation in SOR-CIN through molecular dynamics simulations*. *Journal of Biomolecular Structure and Dynamics*, 42(4), 1812–1825., (2024). https://doi.org/10.1080/07391102.2023.2202772

4. Subramani, M., **Saravanan, V.**, Muralidharan, A., Durai, R., Ramasamy, S. *Theoretical insights on the development of a 55–77 graphene sheet by embedding Agn= 1-4 and Pdn= 1-4 metal nanoclusters for efficient CO2 capture*. *Journal of Molecular Graphics and Modelling*, 124, 108573, (2023). https://doi.org/10.1016/j.jmgm.2023.108573

5. Saravanan, V.,Rajamani, A., Ramasamy, S.,, Baazeem, A., Upadhyaya, IR. *Epigenetically modified nucleobases (5hmc, 5fc, and 5caC) interaction with boron and nitrogen doped porous graphene (B/N-pGr) as promising materials for biosensing application: A density functional theory calculations*. *Environmental Research*, 197, 111133, (2021). https://doi.org/10.1016/j.envres.2021.111133

6. Ravichandran. D., Rajamani, A., **Saravanan, V.**, Ramasamy, S., Manickam, S. *Tuning the reactivity of tri-s-triazine, trinitro-tri-s-triazine and ternary tri-s-triazine graphitic C3N4 quantum dots through H-functionalized and B-doped complexes: A density functional study*. *Chemosphere*, 272, 129901, (2021). https://doi.org/10.1016/j.chemosphere.2021.129901

7. Subramani, M., **Saravanan, V.**, Jayaraman, T., Subramaniam, V., Pazhanivel, T., Ramasamy, S., Manickam S. *Kinetics and degradation of camphene with OH radicals and its subsequent fate under the atmospheric O2 and NO radicals-A theoretical study*. *Chemosphere*, 267, 129250, (2021). https://doi.org/10.1016/j.chemosphere.2020.129250

8. Rajamani, A., **Saravanan, V.**, Subramani, M., Ramasamy, S. *Adsorption of H2 molecules on B/N-doped defected graphene sheets—a DFT study*. *Structural Chemistry*, 6, 2413-2434, (2020). https://doi.org/10.1007/s11224-020-01578-w

9. Rajamani, A., Ravichandran, D., **Saravanan, V.**, Ramasamy, S. *Adsorption of H2 and CO2 gas molecules on Li/Na decorated Si2BN nano-sheet for energy harvesting applications–A density functional study*. *Materials Letters*, 279, 128487, (2020). https://doi.org/10.1016/j.matlet.2020.128487

10. Saravanan, V., Rajamani, A., Subramani, M., Ramasamy, S. *Exploring two-dimensional graphene and boron-nitride as potential nanocarriers for cytarabine and clofarabine anti-cancer drugs*. *Computational Biology and Chemistry*, 88, 107334, (2020). https://doi.org/10.1016/j.compbiolchem.2020.107334

Publications (continued)

11. Saravanan, V., Rajamani, A., Vasudevan, S., Ramasamy, S. *Exploring the nature of interaction and stability between DNA/RNA base pairs and defective and defect-dopant graphene sheets. A possible insights on DNA/RNA sequencing*. *International Journal of Biological Macromolecules*, 146, 387-404, (2020). https://doi.org/10.1016/j.ijbiomac.2020.01.001

12. Saravanan, V., Rajamani, A., Subramaniam, V., Ramasamy, S. *Interaction of (G4)2 and (X4)2 DNA quadruplexes with Cu+, Ag+ and Au+ metal cations: a quantum chemical calculation on structural, energetic and electronic properties*. *Structural Chemistry*, 31, 465–484, (2020). https://doi.org/10.1007/s11224-019-01421-x

13. Rajamani, R., **Saravanan, V.**, Subramani, M., Ramasamy, S. *Reconnoitring the nature of interaction and effect of electric field on Pd/Pt/Ni decorated 5-8-5/55–77 defected graphene sheet for hydrogen storage*. *International Journal of Hydrogen Energy*, 45(1), 744-763, (2020). https://doi.org/10.1016/j.ijhydene.2019.10.170

14. Saravanan, V., Radhika, R., Subramaniam, V., Ramasamy, S. *Structural insights into the anti-cancer activity of quercetin on G-tetrad, mixed G-tetrad, and G-quadruplex DNA using quantum chemical and molecular dynamics simulations*. *Journal of Biomolecular Structure and Dynamics*, 38(2), 317–339, (2019). https://doi.org/10.1080/07391102.2019.1574239

15. Rajamani, R., **Saravanan, V.**, Subramaniam, V., Ramasamy, S. *Modeling of Si–B–N Sheets and Derivatives as a Potential Sorbent Material for the Adsorption of Li+ Ion and CO2 Gas Molecule*. *ACS omega*, 4(9), 13808-13823, (2019). https://doi.org/10.1021/acsomega.9b01354